

deep-research-report_UPSTOX.md

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Summary

This document provides a comprehensive blueprint for developing an **institutional-grade algorithmic trading system** specifically designed for buying NIFTY options via the Upstox API. The system's architecture is built on a **multidisciplinary framework** that divides responsibilities among a Coding Agent for technical integration, a Live Trading Analyst for quantitative research, and an Execution Trader for managing order flow and risk. At its core, the strategy rejects isolated indicators in favor of **composite signals**, correlating complex variables such as implied volatility, gamma exposure, and the put/call ratio to identify high-probability market entries. By mapping out a **phased implementation roadmap** and a rigorous "go/no-go" decision matrix, the report serves as a technical manual for transitioning from raw market data to automated, risk-managed execution during Indian market hours.

Keywords: Options Trading Systems, Upstox API Integration, Quantitative Signal Generation, Risk Management Controls, Algorithmic Order Execution

Content

Executive Summary Building an institutional-grade intraday NIFTY options buying system requires end-to-end integration of market data, quantitative signal generation, and execution components. The system must treat the options ecosystem holistically – correlating implied volatility (IV) skew and trend, time decay (theta), put/call ratio (PCR), gamma exposure (GEX) and market-maker hedging, open interest (OI) walls, and premium VWAP vs. spot VWAP – rather than in isolation. We outline the required skills for each role (Coding Agent, Live Trading Analyst, Execution Trader), map skills to concrete tasks and deliverables, and specify the technical architecture, data pipelines, and algorithms needed. We include example pseudocode and composite signal formulas, an execution "go/no-go" matrix (PCR × IV × GEX × OI regimes), and flow diagrams. The plan assumes Upstox's free APIs (OAuth token, REST and WebSocket market feed) for live NIFTY data (with fallback to

alternative historical sources), and NSE trading hours 9:15–15:30 IST. All development must incorporate robust testing, monitoring, and feedback loops for continuous improvement.

Roles & Responsibilities

Coding Agent: Technical Implementation

API Integration: Implement OAuth2 login and token refresh for Upstox (REST) and maintain a live WebSocket v3 connection. For example, use Upstox’s OAuth 2.0 flow to obtain `access_token` and attach `Authorization: Bearer {token}` to all REST calls. Handle two-factor methods (SMS/TOTP) if needed.

WebSocket Feed (Protobuf): Connect to Upstox Market Data Feed V3 (WSS). Use Protobuf definitions provided by Upstox (e.g. `MarketDataFeed.proto`) to decode real-time messages. Subscribe to `instrumentKeys` for NIFTY spot/future (`NSE_INDEX|Nifty 50` or futures) and relevant options (`NSE_FO|...`). Use modes `ltpc` (LTP/close), `option_greeks`, or `full` as needed. For free accounts, note the subscription limits: up to 5000 keys for LTPC and 3000 for Option Greeks. Implement automatic reconnection and redirection handling in case Upstox reroutes the WSS endpoint.

REST Data Ingestion: Use Upstox REST endpoints to pull option chains and Greeks. For example, call `GET /v2/option/chain?instrument_key=NSE_INDEX|Nifty 50&expiry_date=YYYY-MM-DD` to retrieve strike-wise call/put OI, IV, Greeks, bid/ask etc. Also use `GET /market-quote/quotes` for snapshots or historical candles for backfill.

Data Engineering: Define schemas to store (or stream) fields: timestamps, LTP, bid/ask prices and sizes, volume, OI, IV, delta, gamma, theta, vega, etc. Normalize Upstox field names (e.g. `market_data.oi`→`open_interest`, `option_greeks.iv`→`implied_vol`). For example, Upstox JSON provides: `call_options.market_data.ltp`, `oi`, `bid_price`, and `put_options.option_greeks.iv`. Combine this with spot data. If high-resolution data is needed, also ingest intraday candles via Upstox (V3 intraday endpoints) or store raw ticks from WSS. Ensure all data is timestamped in IST and aligned (e.g. round to nearest second or millisecond).

Fallback Data Sources: Upstox’s free API may not provide historical OI/IV. Plan alternatives: Download daily F&O reports from NSE’s website (which publishes contract-wise OI and volume CSVs) or use public libraries (e.g. `yfinance` or `nsepy`) to fetch historical index futures and options data. Pre-process and reconcile formats (mapping NSE strike/expiry codes to Upstox `instrument_keys`). Document assumptions for missing data (e.g. assume zero OI beyond available history).

Composite Signal Algorithms: Implement the quantitative formulas and pseudocode for the multi-variable signals. For example:

Reference: PCR interpretation and thresholds, VWAP formula, Upstox data fields, and gamma hedging intuition.

Example Code Snippets: Show key patterns:

Error Handling & Rate Limits: Implement retry/backoff for HTTP 4XX/5XX. Respect Upstox REST limits (50 req/sec, 2000/30min) and WS limits. For example, batch `instrument_keys` in chunks (≤ 500 per full-quote call). Handle token expiration by auto-refresh. Log any decode errors on the Protobuf feed.

Unit/Integration Tests & Synthetic Data: Develop unit tests for each data parser (e.g. validate that given a sample JSON/protobuf, the code

extracts correct fields). Create synthetic tick data generators: e.g. simulate an option chain where OI moves or IV expands, to test alert triggers. Example: write tests that check `compute_VWAP()` and `put_call_ratio()` functions return expected values on known inputs.

DevOps & CI/CD: Use version control (Git), automated testing (e.g. pytest), and containerization. Continuous integration should run tests on each commit. Deployment checklist: code linting, config for tokens (securely handled), SSL for WSS.

Live Trading Analyst: Quant Research & Monitoring

Quantitative Modeling: Deep understanding of options Greeks, volatility surfaces, and structural flows. Skills in statistics/Python/R for backtesting signals. Familiarity with options theory (e.g. volatility skew, forward PDE for decay).

Data Analysis: Monitor real-time signals (PCR, IV skew, premium flows) and verify against market outcomes. For example, track how PCR regimes (<0.7 , $0.7-1.1$, $1.1-1.4$) have historically correlated with NIFTY moves. Use tools (Python, Pandas) to compute moving averages/slopes of PCR and IV.

Composite Signal Integration: Implement and refine composite triggers. For instance, code a "Momentum Index" as: $M = w_1 + w_2 (\text{VWAP}_{\text{call}} - \text{VWAP}_{\text{spot}}) + w_3$, where weights (w_i) are tuned. Use pseudocode (see Coding Agent) to prototype. Validate signals by backtesting on historical intraday data.

Strategy & Risk Definition: Define explicit GO/NO-GO rules. E.g., flag a "no-trade" scenario when $\text{PCR} \approx 1$, Max Pain level is static, and large straddle OI builds (institution writing strangles). Conversely, identify "gamma-hunt" buy setups: heavy call walls, spot near zero-gamma point, short gamma exposure (see Composite Matrix).

Backtesting & Metrics: Build a backtest framework to simulate trades. Key metrics: hit-rate, average gain/loss, drawdown, Sharpe ratio, latency impact. For example, simulate buying ATM NIFTY calls when our composite conditions are met and track P/L. Use non-overfitting practices (e.g. walk-forward).

Monitoring & Reporting: Implement real-time dashboards. Plot live PCR, IV, GEX, P&L. Alert on signal events. Continuously record system performance and P&L for calibration. Example visual: PCR vs. time of day plot, VWAP premium vs. time chart. Document every anomaly or false signal and iterate.

Execution Trader: Order Execution & Risk Controls

Market Microstructure: Expertise in NSE F&O specifics: trading hours (9:15–15:30 IST), tick size (₹0.05 for index options), lot sizes, and typical intraday patterns (U-shaped volume/spread). Understand how bid-ask spreads widen intraday (widest at open/close) and thus tailor execution.

Order Types & Slicing: Use smart order execution. E.g., break a large call-buy order into smaller limit orders ("iceberg" algorithm) to reduce slippage. Decide order type by liquidity: for highly liquid near-ATM calls use Aggressive limit (or IOC), for thin far-OTM use cautious. Incorporate delta-hedge triggers: e.g., once a call position is entered, hedge by selling futures when delta exceeds threshold.

Execution Algorithms: Pseudocode example: Use real-time premium VWAP vs. spot VWAP to gauge if buying is populating at premium or mid-market. If premium/VWAP divergence is large, be prepared for adverse fill and widen limits.

Monitor execution slippage and adjust size thresholds (e.g. reduce slice size if fills are slow).

Risk Management: Enforce stop-loss and position limits. If a trade goes against the signal (e.g. price moves opposite despite positive signal), cut losses quickly. For multiple simultaneous edges, allocate capital proportionally. During high-IV periods (which inflate premiums), reduce notional to control Vega risk.

Time-of-Day Filters: Different tactics by session. For example:

- Morning (9:15–10:15):** Use first-hour momentum; be aggressive on clear signals (higher liquidity, tighter spreads).
- Midday (11:30–13:30):** Markets lull; lean passive. Only trade high-conviction signals (perhaps smaller size).
- Afternoon (14:00–15:30):** Watch for expiry/position squaring. Leverage any shift in Max Pain or large final hour flows to enter/exit. For instance, if Max Pain shifts up and PCR rises late, consider late entry. If near expiry and a big gamma pin is expected, be ready to sell or close out.

GO/NO-GO Decision Rules: Implement a simple decision table (illustrated below) per time block. E.g., **“No-Trade”** if $PCR \approx 1$ and IV flat and large adjacent CE/PE OI added simultaneously – indicates a big straddle write (time decay ramp) and premiums draining. **“Trade”** if PCR rises from <0.7 , IV expanding, and a large underlying move is initiated, signalling momentum. These rules translate to clear intraday trading actions or flat positions.

Skill-to-Task Mapping

Composite Execution Matrix This table summarizes ideal **Go/No-Go** triggers combining PCR, IV trend, GEX and OI patterns. (Cells below describe typical scenarios; actual thresholds to be calibrated.)

Go: Proceed with a trade; consider position size. **No-Go:** Avoid initiating new buys (or flatten existing positions).

This matrix is a guideline. Actual triggers use continuous measurements (e.g. PCR slope, IV skew rate, distance to zero-gamma). E.g. a high PCR in row 3 combined with a sudden spike in spot moving through a call wall can flip to **Go** (covering short puts). The “No-Go” signature (row 5) reflects $PCR \approx 1.0$, static Max Pain, and simultaneous big OI adds on both calls and puts – a classic straddle-write trap (time decay heavy).

Example Algorithms & Pseudocode

- Composite Momentum Signal** – e.g. an index combining PCR trend, VWAP divergence, and delta-change.
- Gamma Hedge Trigger** – anticipate market-maker hedging.
- Order Slicing Algorithm** – microstructure-aware execution.

VWAP Calculation (for illustration):

$$[VWAP = P_i, V_i]$$

Data Flow & Time-of-Day Diagrams Below are schematic diagrams of the system data flow and intraday time segmentation. (When exporting to PDF or DOC, convert these Mermaid diagrams to images. For example, generate the diagram PNG and embed as

Data Flow Diagram

.)Implementation Roadmap

Phase 1 – Data & API Setup (Weeks 1–2): Obtain Upstox API keys. Implement login/auth module (OAuth flow). Connect to WebSocket feed and REST to confirm data access. Parse sample data (option chain, quotes) and verify fields. Establish database or message queue schema. **Milestone:** Real-time

data pipeline up and running (with logging).Phase 2 – Core Metric Computation (Weeks 2–4): Code routines for PCR, IV skew, theta acceleration, net gamma, premium VWAP, etc. Ingest a few days of historical NIFTY options data (via Upstox or alternative) and compute these metrics offline. **Milestone:** Standalone backtest scripts producing historical signal plots (e.g. PCR time series vs. index moves).Phase 3 – Signal Integration & Backtest (Weeks 4–6): Combine metrics into composite signals (per pseudocode above). Implement signal engine that scans live feed and flags conditions. Backtest these signals on historical intraday data to estimate performance (P&L, hit rate, drawdown). **Milestone:** Documented performance report; adjustable parameters for signal thresholds.Phase 4 – Execution Module (Weeks 6–8): Build order execution logic. Simulate order placements with hypothetical fills to estimate slippage. Integrate delta-hedge routine (link option buys to underlying futures sells). **Milestone:** End-to-end simulated trading from signal to order logic, with risk checks (size limits, stops).Phase 5 – Deployment & Monitoring (Weeks 8–10): Containerize system, deploy to a cloud server or VPS. Set up continuous integration (GitHub Actions) and monitoring dashboards (e.g. Grafana for latency, logs). Develop alerting (email/SMS) for signals and errors. Document security (store API keys securely, refresh tokens). **Milestone:** Live paper-trading or small real-money run with logging of all metrics (latency, fill quality, P&L).Phase 6 – Iteration & Learning (Ongoing): Collect trading outcomes and label each trade (win/lose/cancel). Analysts refine signal models using real performance feedback. Implement drift detection on input features (if PCR/IV distributions shift, retrain thresholds). Continually incorporate new data (e.g. annual NSE contract changes) and update the system.Each phase should include peer reviews and regression testing. Progress along this roadmap ensures readiness for live deployment under strict risk controls.

Sources: Upstox API documentation provides the exact fields for market data and option chain (e.g. market_data.oi, option_greeks.iv). Real-time feed limits are noted in Upstox docs. Market-microstructure context (U-shaped intraday spreads) is from NSE research. Put/Call Ratio interpretation is guided by Investopedia. VWAP formula is standard, and gamma exposure impacts are described in quantitative trading literature. NSE trading hours (9:15–15:30 IST) are from official schedule. Each of these underpins the design above.